ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** growth rates vary and the cousin may have started his spurt	Individuals grow differently.
2	**(a)** larynx is growing and adjusting	Settles in time.
3	**(a)** a rapid rise during the growth spurt	Sudden increase.
4	**(c)** it is within the normal range	Normal variation.
5	**(c)** menstruation	Ovary hormones needed.
6	**(b)** around 11 to 13 years	Growth spurt.
7	**(a)** anaemia	Iron lack.
8	**(d)** be affected over time	Hormone loss.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Growth spurts and other changes of the teenage years are brought about by hormones.
10	**(c)** A is true but R is false.	The pituitary controls other glands, so A is true; insulin is made by the pancreas, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hormones carry messages through blood (1); They control growth and mood (1).	1+1
12	Testosterone promotes muscle growth (1); Greater rise in boys (1).	1+1
13	Hormonal changes affect mood (1); Support prevents stress (1).	1+1
14	Heredity and nutrition differ (1); Normal variation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Growth hormone lacking (1); Poor growth (1); Other glands underactive (1).	1+1+1
16	Avoid excess sweets (1); Carry medicine (1); Eat at regular times (1).	1+1+1
17	Alertness (1); Energy (1); But excess causes stress (1).	1+1+1
	OR: Vitamins deficiency (1); Add fruits and vegetables (1); Reduce fried food (1).	1+1+1
18	Sunlight gives vitamin D (1); Helps calcium absorption (1); Strong bones (1).	1+1+1
	OR: Stand up politely (1); Explain it is natural (1); Inform teacher (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Puberty (adolescence) (1). (ii) Testosterone (1). (iii) The testes (1). (iv) The voice box (larynx) grows larger at puberty (1), so the voice becomes deeper; in boys the enlarged voice box can be seen as the Adam's apple (1).	1+1+1+2
20	Nerves: fast signals (1); Hormones: slower, lasting (1); Adrenaline for stress (1); Sex hormones for puberty (1); Both coordinate (1).	1+1+1+1+1
	OR: Poor diet (1); Low vitamins and iron (1); Excess oil and sugar (1); Better diet (1); Exercise and hygiene (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
- Writing that hormones are carried by ducts; endocrine glands are **ductless** and hormones travel in the blood.
- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** rise	No control.
2	**(c)** common as cycles settle	Cycles take time.
3	**(c)** insulin – pancreas	Insulin is from the pancreas.
4	**(a)** sugar	Reduce sugar.
5	**(b)** inform a trusted adult	Seek help.
6	**(d)** gland, blood, target organ	Blood transport.
7	**(b)** adrenaline	Emergency.
8	**(a)** body is not fully ready	Health risks.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but one fact does not explain the other.
10	**(d)** A is false but R is true.	Endocrine glands have no ducts and release hormones into the blood, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sweat glands are active (1); Regular bathing and clean clothes (1).	1+1
12	Uterine muscles contract (1); Hormone changes (1).	1+1
13	It secretes hormones (1); These hormones control other glands (1).	1+1
14	Body needs trace amounts (1); Excess can also harm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Insulin injection in diabetes (1); Replaces missing hormone (1); Regular monitoring (1).	1+1+1
16	Family patterns (1); Inherited traits affect growth (1); Time varies (1).	1+1+1
17	Set small goals (1); Learn skills (1); Accept self (1).	1+1+1
	OR: Primary: testes, ovaries (1); Secondary: beard, breasts (1); Secondary appear at puberty (1).	1+1+1
18	Late-night usage (1); Sleep loss (1); Poor concentration (1).	1+1+1
	OR: Strong bones (1); Healthy weight (1); Fewer diseases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Menstruation (1). (ii) The ovary (1). (iii) About every 28 to 30 days (1). (iv) Blood is lost during menstruation and iron helps to make haemoglobin in red blood cells (1); clean pads or cloth and washing prevent infections (1).	1+1+1+2
20	Pituitary controls thyroid (1); Thyroid controls metabolism (1); Pancreas controls sugar (1); Failure: goitre or diabetes (1); Treatment (1).	1+1+1+1+1
	OR: Poor growth (1); Anaemia (1); Mid-day meal (1); Iron supplements (1); Health camps (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
- Writing that hormones are carried by ducts; endocrine glands are **ductless** and hormones travel in the blood.
- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** do not share personal details with strangers	Safety.
2	**(b)** endocrine glands pour hormones directly into blood	Ductless glands.
3	**(d)** dal, rice, vegetables, curd and fruit	Balanced.
4	**(c)** maturity also needs emotional and mental growth	Social and emotional maturity.
5	**(a)** secretion of insulin	Insulin is from the pancreas.
6	**(c)** screen use and heavy caffeine	Disturb sleep.
7	**(c)** assertive communication	Firmly and kindly say no.
8	**(b)** think about the risk and say no	Safe decision.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Fast growth increases the need for nutrients, which explains the need for a richer diet.
10	**(a)** Both A and R are true and R is the correct explanation of A.	A larger voice box produces a deeper voice, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Prevents girls from seeking information (1); Affects confidence and health (1).	1+1
12	Rapid bone growth (1); Blood formation in menstruation (1).	1+1
13	Frequent urination and thirst (1); Doctor's test of blood sugar (1).	1+1
14	Reduces stress (1); Improves mood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Body grows (1); Mind develops (1); Responsibilities increase (1).	1+1+1
16	Reduces misunderstanding (1); Solves problem (1); Builds trust (1).	1+1+1
17	Prevent anaemia (1); Blood loss in periods (1); Better learning (1).	1+1+1
	OR: Health talks (1); Counselling (1); Sports and nutrition programmes (1).	1+1+1
18	Menstruation is impure: false, natural (1); Voice cracking is a disease: false (1); Pimples due to dirt only: false, hormones (1).	1+1+1
	OR: Safe space (1); Expert guidance (1); Helps solve problems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A diet containing all nutrients in the right amounts (1). (ii) Hormones (1). (iii) Proteins (or calcium or iron) (1). (iv) Exercise and enough sleep (1); they keep the body fit, help growth and help control emotions and mood changes (1).	1+1+1+2
20	Iron: spinach, jaggery, dates (1); Calcium: milk, curd (1); Protein: dal, eggs (1); Hygiene: daily bath, clean clothes (1); Water and exercise (1).	1+1+1+1+1
	OR: Study seriously (1); Respect others (1); Take care of health (1); Follow laws (1); Help at home (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
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